

EVENING

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[Total No. of Questions: 09]

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Uni. Roll No.

Program: B.Tech. (Batch 2018 onward)

Semester: 3rd

Name of Subject: Data Structures

Subject Code: PCIT-101

Paper ID: 16040

Scientific calculator is Allowed

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice.
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q. 1

- a) Describe the Big Oh Notation.
- b) Interpret the efficiency of Binary Search algorithm.
- c) Illustrate the concept of Priority Queue.
- d) Discuss Circular Linked List.
- e) Compare adjacency matrix and adjacency list.
- f) Appraise the importance of rehashing.

Part – B

[Marks: 04 each]

- Q. 2** Demonstrate the working of quicksort algorithm.
- Q. 3** Describe the preorder, inorder and postorder traversals of a binary tree.
- Q. 4** Discuss the use of stack data structure in solving Tower of Hanoi problem.
- Q. 5** Compare the efficiency of linear queue and circular queue.
- Q. 6** Support the statement that an AVL tree is a self-balancing binary search tree.
- Q. 7** Judge the statement that the threaded binary trees enhances the traversal techniques by improving its speed.

Part – C

[Marks: 12 each]

- Q. 8** Demonstrate the operations of insertion and deletion in a linear linked list.

OR

Demonstrate how stack data structure is used in the conversion of infix to postfix expression.

- Q.9** Compare the Breadth First Search (BFS) and Depth First Search (DFS) graph traversal techniques.

OR

Defend the statement that hashing is a technique which modifies the given key to a smaller value so that a great deal of space is not wasted.
